

## IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY &amp; MEDICINE

## Department of Aeronautics

## Aircraft Aerodynamics

## Part III

## Tutorial Sheet 1

(Questions 1 - 4 are for revision)

1. In a two-dimensional velocity field

$$u = 4xy + y^2$$

$$v = -2y^2 + 3x^2 + 4.$$

Do these velocity components satisfy continuity?

Determine the vorticity in this flow.

Ans.  $2(x-y)$

*For continuity,*

$$\nabla \cdot \mathbf{U} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 4y - 4y = 0$$

*$\therefore$  satisfy continuity*

$$\omega = \nabla \times \mathbf{U} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 6x - (4x + 2y) = 2x - 2y$$

2. The velocity potentials of two flows are given by

$$\phi_1 = -4 \log_e r \text{ and } \phi_2 = 6x.$$

If the two flows are superimposed, what is the stream function of the resulting combined flow? Sketch some streamlines and show that the flow corresponds to that near the rounded trailing edge of a semi-infinite body.

$$\text{Ans. } 6y - 4 \tan^{-1} y/x + \text{Constant.}$$

$$* \log_e = \ln$$

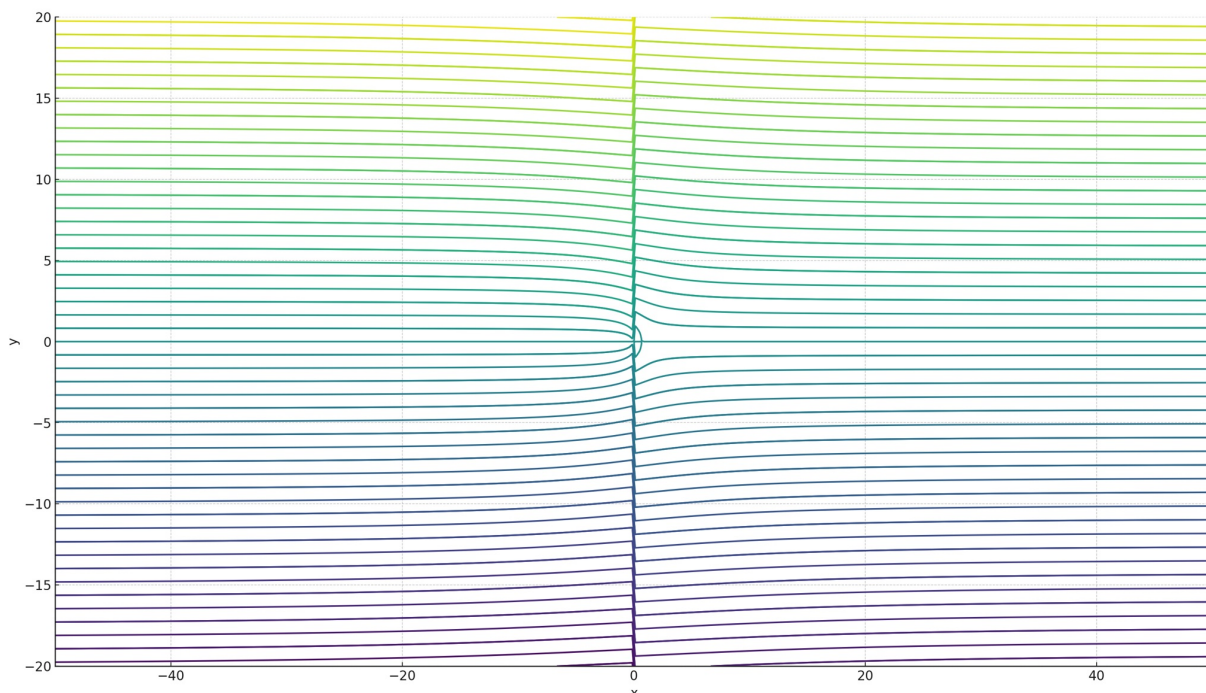
$$u = \frac{\partial \psi}{\partial y} = \frac{\partial \phi}{\partial x}, \quad v = -\frac{\partial \psi}{\partial x} = \frac{\partial \phi}{\partial y}$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{\partial \phi}{\partial r}, \quad V_\theta = -\frac{\partial \psi}{\partial r} = \frac{1}{r} \frac{\partial \phi}{\partial \theta}$$

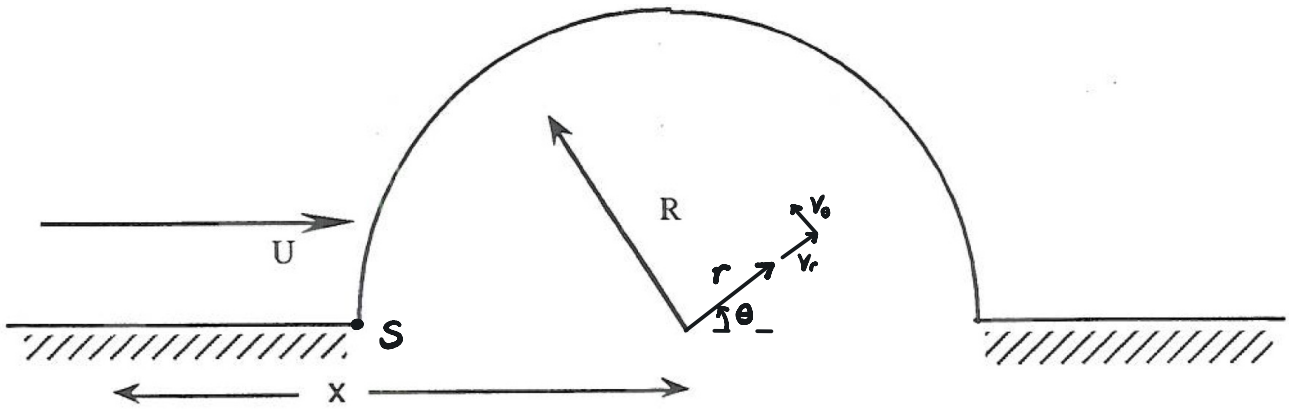
$$\text{for } \phi_1, \quad \frac{1}{r} \frac{\partial \psi_1}{\partial \theta} = -\frac{4}{r} \quad \text{and} \quad -\frac{\partial \psi_1}{\partial r} = 0 \quad \therefore \psi_1 = -4\theta + C_1$$

$$\text{for } \phi_2, \quad \frac{\partial \psi_2}{\partial y} = 6 \quad \text{and} \quad -\frac{\partial \psi_2}{\partial x} = 0 \quad \therefore \psi_2 = 6y + C_2$$

$$\therefore \psi = \psi_1 + \psi_2 = -4\theta + 6y + C = 6y - 4 \tan^{-1} \left( \frac{y}{x} \right) + C$$



3. An incompressible inviscid fluid is bounded by the surface whose section is shown in the sketch. Far upstream from the cylindrical hump, the fluid flows in uniform motion parallel to the surface. Find the pressure distribution on the plane surface in the vicinity of the hump, according to potential flow theory.



What changes in the flow pattern would you expect if a real fluid were flowing over the same surface?

Ans.  $1 - (1 - R^2/x^2)^2$

$$\psi = \psi_{\text{velocity}} + \psi_{\text{circulation}}$$

$$\text{For } \psi_{\text{velocity}}, (V_\theta)_v = -\frac{\partial \psi_v}{\partial r} = -U \sin \theta \quad \therefore \psi_v = Ur \sin \theta + C_1$$

$$\therefore \psi = Ur \sin \theta - \frac{k}{2\pi r} \sin \theta \quad (0 \leq \theta \leq \pi)$$

At the surface of hump,  $r = R$ ,  $V_r = 0$

$$\therefore V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U \cos \theta - \frac{k}{2\pi R^2} \cos \theta = 0 \quad \therefore k = 2\pi R^2 U$$

$$\text{Therefore, } \psi = U \sin \theta \left( r - \frac{R^2}{r} \right)$$

$$\therefore V_\theta = -\frac{\partial \psi}{\partial r} = -U \sin \theta \left( 1 + \frac{R^2}{r^2} \right)$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U \cos \theta \left( 1 - \frac{R^2}{r^2} \right)$$

On the plane surface,  $r = x$  and  $\theta = \pi$

$$\therefore V_\theta = 0 \text{ and } V_r = -U \left( 1 - \frac{R^2}{x^2} \right)$$

According to Bernoulli equation,

$$p_0 = p_\infty + \frac{1}{2}\rho U_\infty^2 = p + \frac{1}{2}\rho U^2$$

$$\therefore C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2} = 1 - \left(\frac{U}{U_\infty}\right)^2$$

$$= 1 - \left[ \frac{-U \left(1 - \frac{R^2}{x^2}\right)}{U} \right]^2$$

$$= 1 - \left(1 - \frac{R^2}{x^2}\right)^2$$



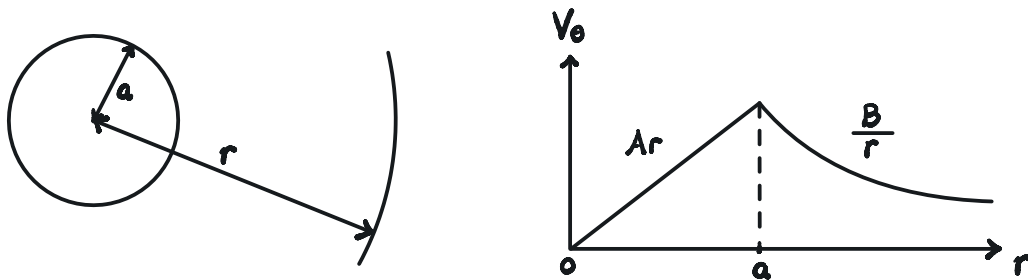
4. A vortex has a central core of radius,  $a$ , in which the tangential velocity increases linearly with radius. Outside the core the tangential velocity at any radius  $r$  can be written as  $u_\theta = B/r$ , where  $B$  is a constant. Show that the circulation  $\Gamma$  at any radius  $r$  inside the core is

$$\Gamma = 2\pi Br^2/a^2.$$

If the pressure at infinity is  $p_\infty$  show that the pressure at the centre of the core is

$$p_\infty - \rho B^2/a^2,$$

where  $\rho$  is the density



$$\therefore Ar = \frac{B}{r} \text{ at } r = a \quad \therefore A = \frac{B}{a^2}$$

$$\Gamma = \int U dl$$

$$= 2\pi r \cdot V_\theta = 2\pi r \cdot (Ar) = 2\pi r^2 \frac{B}{a^2}$$

According to Bernoulli equation,

$$p_0 = p_\infty + \frac{1}{2}\rho U_\infty^2 = p + \frac{1}{2}\rho U^2$$

$$\therefore p_\infty + \frac{1}{2}\rho \left(\frac{B}{r}\right)^2 \Big|_{r \rightarrow \infty} = p + \frac{1}{2}\rho \left(\frac{B}{a^2}r\right)^2$$

At centre of core,  $r = a$

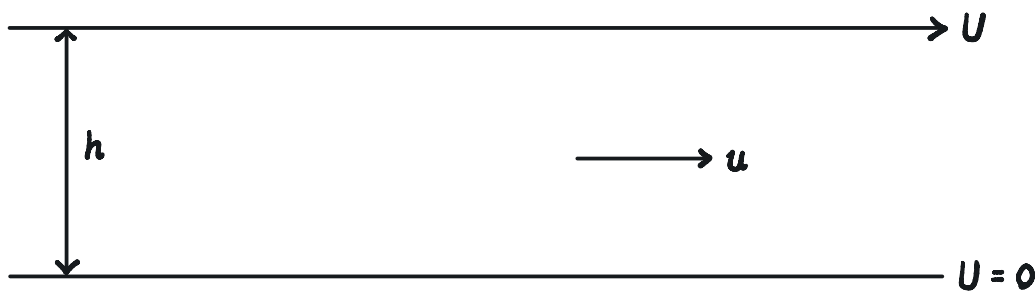
$$p_\infty = p + \frac{1}{2}\rho \left(\frac{B}{a}\right)^2$$

$$\therefore p = p_\infty - \frac{1}{2}\rho \frac{B^2}{a^2}$$

5. A fluid is confined between two infinite parallel flat plates separated by a distance  $h$ . One plate is stationary, while the other is moving at velocity  $U$  in the  $x$ -direction. There is a pressure gradient in the  $x$  direction,  $dp/dx$ .

Starting from the equations of motion, show that the local velocity,  $u$ , is given by

$$\frac{u}{U} = \frac{y}{h} \left( 1 - \frac{h^2}{2\mu U} \frac{dp}{dx} \left( 1 - \frac{y}{h} \right) \right)$$



Apply  $x$ -component of RANS equation for laminar flow, since  $\frac{b}{x} \ll 1$

$$U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 U}{\partial y^2}$$

since  $\frac{\partial U}{\partial x} = V = 0$  between parallel plates,

$$\nu \frac{\partial^2 U}{\partial y^2} = \frac{1}{\rho} \frac{dp}{dx} \quad \therefore \frac{\partial^2 U}{\partial y^2} = \frac{1}{\mu} \frac{dp}{dx}$$

$$\therefore u = \frac{1}{2\mu} \frac{dp}{dx} y^2 + Cy + D$$

$$\text{when } y = 0, u = 0 \quad \therefore D = 0$$

$$\text{when } y = h, u = U \quad \therefore \frac{1}{2\mu} \frac{dp}{dx} h^2 + Ch = U$$

$$\therefore C = \frac{U}{h} - \frac{1}{2\mu} \frac{dp}{dx} h$$

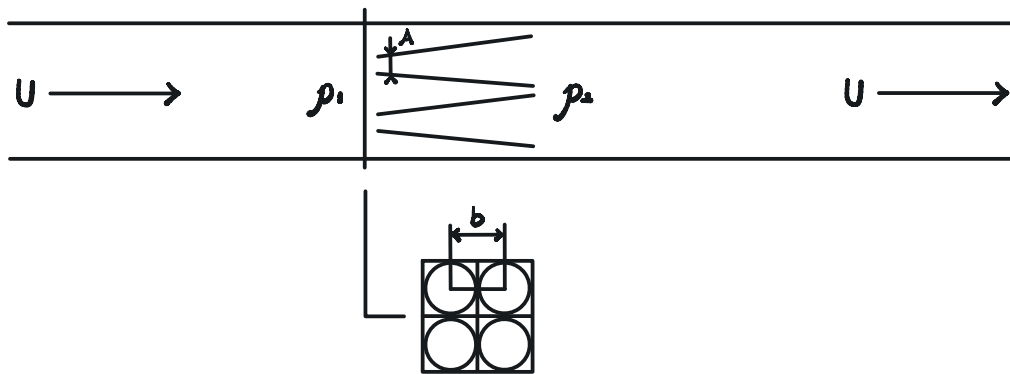
$$\therefore u = \frac{1}{2\mu} \frac{dp}{dx} y^2 + \left( \frac{U}{h} - \frac{1}{2\mu} \frac{dp}{dx} h \right) y$$

$$\therefore \frac{u}{U} = \frac{1}{2\mu} \frac{dp}{dx} \frac{1}{U} y^2 + \frac{y}{h} - \frac{1}{2\mu} \frac{dp}{dx} y h = \frac{y}{h} \left[ 1 - \frac{h^2}{2\mu U} \frac{dp}{dx} \left( 1 - \frac{y}{h} \right) \right]$$

6. Incompressible fluid of density  $\rho$  flows along a straight duct of uniform cross-section. A flat perforated plate extends across the duct, normal to the direction of flow. The velocity far upstream of the plate is  $U$ , and the plate is uniformly perforated in a square pattern by circular holes with centres spaced a distance,  $b$  apart. The flow issues from the plate as a series of circular jets, for which the minimum area is  $A$ . Further downstream the flow mixes until the speed is again  $U$ , while the static pressure is less than that upstream by an amount  $\Delta p$ . Show that

$$\frac{\Delta p}{\frac{1}{2}\rho U^2} = \left( \frac{b^2}{A} - 1 \right)^2$$

and state any assumptions you have made.



Assume the flow is inviscid and irrotational. apply Bernoulli:

$$p_1 + \frac{1}{2}\rho U^2 = p_2 + \frac{1}{2}\rho U_2^2$$

$$\therefore \Delta p = |p_1 - p_2| = \frac{1}{2}\rho U_2^2 - \frac{1}{2}\rho U^2 = \frac{1}{2}\rho U^2 \left( \frac{U_2^2}{U^2} - 1 \right) \quad \therefore \frac{\Delta p}{\frac{1}{2}\rho U^2} = \frac{U_2^2}{U^2} - 1$$

Assume there are  $N$  holes on the plate.  $A_{\text{plate}} = Nb^2$

Therefore, apply continuity,  $\rho U A_{\text{plate}} = \rho U_2 N A$

$$\therefore U N b^2 = U_2 N A \quad \therefore U_2 = U \frac{b^2}{A}$$

$$\therefore \frac{\Delta p}{\frac{1}{2}\rho U^2} = \left( \frac{b^2}{A} \right)^2 - 1$$

7. A line source of strength  $m$  is placed a distance  $h$  from an infinite flat wall. If the pressure on the other side of the wall is equal to that at infinity, show that the force experienced by the wall is  $\rho m^2 / 4\pi h$ . (The line source is of course, perpendicular to the line made by taking a section through the wall.)

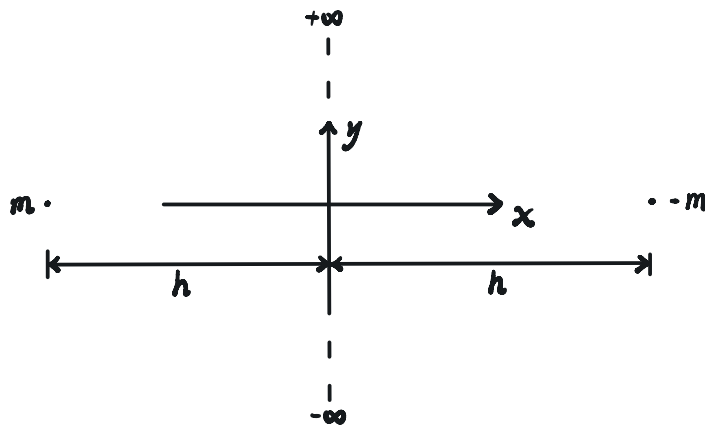
Assume incompressible, irrotational and inviscid flow

$$V_r = \frac{m}{2\pi r}$$

$$\therefore V_r = \frac{d\phi}{dr}$$

$$\therefore \phi = \frac{m}{2\pi} \ln r$$

Apply method of images, take cartesian coordinate as shown below



$$\text{Therefore, } \phi = \phi_m + \phi_{-m} = \frac{m}{2\pi} \ln \sqrt{(x+h)^2 + y^2} - \frac{m}{2\pi} \ln \sqrt{(x-h)^2 + y^2}$$

$$\therefore u_m = \frac{d\phi_m}{dx} = \frac{m}{2\pi} \frac{(x+h)[(x+h)^2 + y^2]^{-\frac{1}{2}}}{\sqrt{(x+h)^2 + y^2}} = \frac{m}{2\pi} \frac{x+h}{(x+h)^2 + y^2}$$

$$u_{-m} = \frac{d\phi_{-m}}{dx} = -\frac{m}{2\pi} \frac{x-h}{(x-h)^2 + y^2}$$

$$u = u_m + u_{-m} = \frac{m}{2\pi} \left[ \frac{x+h}{(x+h)^2 + y^2} - \frac{x-h}{(x-h)^2 + y^2} \right]$$

$$\text{similarly, } v = \frac{m}{2\pi} \left[ \frac{y}{(x+h)^2 + y^2} - \frac{y}{(x-h)^2 + y^2} \right]$$

At wall,  $x = 0$ ,

$$u = \frac{m}{\pi} \frac{h}{y^2 + h^2}, \quad v = 0$$

According to Bernoulli,  $p_{\infty} + \frac{1}{2}\rho \cdot 0 = p + \frac{1}{2}\rho U^2 = p + \frac{1}{2}\rho u^2$

$$\therefore F = \Delta p A$$

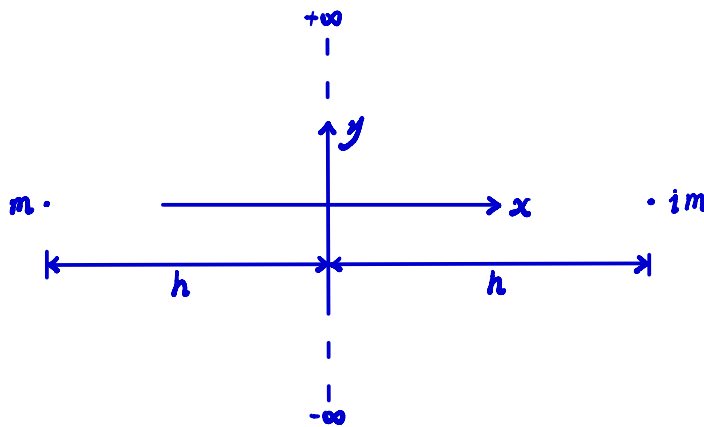
$$\therefore F = \int \Delta p \, dy = \int_{-\infty}^{\infty} \frac{1}{2}\rho \frac{m^2}{\pi^2} \frac{h^2}{(y^2+h^2)^2} \, dy$$

$$= \frac{1}{2}\rho \frac{m^2}{\pi^2} h^2 \int_{-\infty}^{\infty} \frac{1}{(y^2+h^2)^2} \, dy$$

$$= \frac{\rho m^2 h^2}{2\pi^2} \cdot \frac{\pi}{2h^3}$$

$$= \frac{\rho m^2}{4\pi h}$$

Apply method of images, take cartesian coordinate as shown below



$$\text{Therefore, } \phi = \phi_m + \phi_{-m} = \frac{m}{2\pi} \ln \sqrt{(x+h)^2 + y^2} + \frac{m}{2\pi} \ln \sqrt{(x-h)^2 + y^2}$$

$$\therefore u_m = \frac{d\phi_m}{dx} = \frac{m}{2\pi} \frac{(x+h)[(x+h)^2 + y^2]^{-\frac{1}{2}}}{\sqrt{(x+h)^2 + y^2}} = \frac{m}{2\pi} \frac{x+h}{(x+h)^2 + y^2}$$

$$u_{-m} = \frac{d\phi_{-m}}{dx} = \frac{m}{2\pi} \frac{x-h}{(x-h)^2 + y^2}$$

$$u = u_m + u_{-m} = \frac{m}{2\pi} \left[ \frac{x+h}{(x+h)^2 + y^2} + \frac{x-h}{(x-h)^2 + y^2} \right]$$

$$\text{similarly, } v = \frac{m}{2\pi} \left[ \frac{y}{(x+h)^2 + y^2} + \frac{y}{(x-h)^2 + y^2} \right]$$

At wall,  $x = 0$ ,

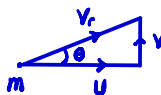
$$u = 0, \quad v = \frac{m}{\pi} \frac{y}{y^2 + h^2}$$

According to Bernoulli,  $p + \frac{1}{2}\rho U^2 = p_{w-} = p_{w+} + \frac{1}{2}\rho v^2$

$$\begin{aligned}\therefore F &= \Delta p A = \int \frac{1}{2}\rho v^2 dy \\&= \frac{\rho m^2}{2\pi^2} \int_{-\infty}^{\infty} \frac{y^2}{(y^2+h^2)^2} dy \\&= \frac{\rho m^2}{2\pi^2} \int_{-\infty}^{\infty} \frac{1}{y^2+h^2} - \frac{h^2}{(y^2+h^2)^2} dy \\&= \frac{\rho m^2}{2\pi^2} \cdot \frac{\pi}{2h} \\&= \frac{\rho m^2}{4\pi h}\end{aligned}$$

OR

for line source,  $U_{total} = 0$  at wall



$$\therefore V_{wall} = V_r|_{wall} \cdot \sin\theta = \frac{m}{2\pi r} \frac{y}{r} \quad \text{where } r = \sqrt{y^2+h^2}$$

$$\therefore V_{total}|_{wall} = 2V_{wall} = \frac{my}{\pi r^2}$$

$$\begin{aligned}\therefore F_{wall} &= \Delta p \cdot \text{thickness} = \int_{-\infty}^{\infty} \Delta p \cdot dy = \int_{-\infty}^{\infty} \frac{1}{2}\rho U^2 dy = \int_{-\infty}^{\infty} \frac{1}{2}\rho V_{total}|_{wall}^2 dy \\&= \frac{\rho m^2}{2\pi^2} \int_{-\infty}^{\infty} \frac{y^2}{r^4} dy\end{aligned}$$

$$\begin{aligned}\text{where } \int_{-\infty}^{\infty} \frac{y^2}{r^4} dy &= \int_{-\infty}^{\infty} \frac{y^2}{(y^2+h^2)^2} dy = \int_{-\infty}^{\infty} \frac{y^2+h^2-h^2}{(y^2+h^2)^2} dy \\&= \int_{-\infty}^{\infty} \frac{1}{y^2+h^2} dy - \int_{-\infty}^{\infty} \frac{h^2}{(y^2+h^2)^2} dy = \int_{-\infty}^{\infty} \frac{1}{y^2+h^2} dy - h^2 \int_{-\infty}^{\infty} \frac{1}{(y^2+h^2)^2} dy\end{aligned}$$

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8. Starting from the Navier-Stokes equations, derive Bernoulli's equation,  $\frac{p}{\rho} + \frac{U^2}{2} + gh \equiv \text{constant}$ , for a steady, irrotational, incompressible flow field where

$$U^2 = U^2 + V^2 + W^2.$$

You should assume that a body force,  $F_y = -\rho g$ , acts in the y-component equation.

N-S equation provides that,

$$\frac{DU_i}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \nabla^2 U_i + \frac{1}{\rho} F_i$$

for a steady, incompressible flow

$$U \frac{\partial U_i}{\partial x} + V \frac{\partial U_i}{\partial y} + W \frac{\partial U_i}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{1}{\rho} F_i$$

therefore,

$$\begin{cases} U \frac{\partial U}{\partial x} + V \frac{\partial U}{\partial y} + W \frac{\partial U}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \\ U \frac{\partial V}{\partial x} + V \frac{\partial V}{\partial y} + W \frac{\partial V}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} - g \\ U \frac{\partial W}{\partial x} + V \frac{\partial W}{\partial y} + W \frac{\partial W}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} \end{cases}$$

for irrotational flow,

$$\omega = \nabla \times U = 0$$

$$\therefore \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ U & V & W \end{vmatrix} = 0 \quad \therefore \begin{cases} \frac{\partial W}{\partial y} - \frac{\partial V}{\partial z} = 0 \\ \frac{\partial U}{\partial z} - \frac{\partial W}{\partial x} = 0 \\ \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} = 0 \end{cases}$$

x, z

substitute into y-component equation, get

$$U \left( \frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z} \right) + V \left( \frac{\partial V}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial V}{\partial z} \right) + W \left( \frac{\partial W}{\partial x} + \frac{\partial W}{\partial y} + \frac{\partial W}{\partial z} \right) = -\frac{1}{\rho} \left( \frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} + \frac{\partial p}{\partial z} \right) - g$$

$$\therefore \boxed{\frac{1}{2} \frac{\partial}{\partial y} (U^2 + V^2 + W^2) = -\frac{1}{\rho} \frac{\partial p}{\partial y} - g} \quad \frac{1}{2} \nabla U^2 = -\frac{1}{\rho} \nabla p - g$$

$$\therefore \frac{1}{2} U^2 = -\frac{p}{\rho} - gh + C$$

$$\therefore \frac{p}{\rho} + \frac{U^2}{2} + gh = C$$